

Python Basics

Python is a simple, high-level programming language used for web development, data science, automation, AI, and scripting. It is beginner-friendly because its syntax is easy to read.

Features

- Easy syntax
- Interpreted language
- Object-oriented
- Large standard library
- Cross-platform

First Program

```
print('Hello, World!')
```

Variables and Data Types

Variables store values in memory.

```
name= 'Taseen '  
age=21  
height=5.4  
is_student=True
```

Data Types

- int – whole numbers
- float – decimal numbers
- str – text
- bool – True/False
- list – collection of items

Operators and Conditions

Arithmetic Operators

```
a=10; b=3  
print(a+b, a-b, a*b, a/b, a%b)
```

Comparison Operators

```
==, !=, {'>'}, {'<'}, {'>='}, {'<='}
```

If Statement

```
age=18  
if age>=18:  
    print('Adult')  
else:  
    print('Minor')
```

Loops and Functions

For Loop

```
for i in range(1,6):  
    print(i)
```

While Loop

```
count=1  
while count<=3:  
    print(count)  
    count+=1
```

Function

```
def greet(name):  
    return 'Hello '+name
```

Lists, Dictionaries & Practice

List

```
fruits=['Apple','Mango','Banana']
```

Dictionary

```
student={'name':'Taseen','marks':95}
```

Practice Questions

1. What is Python?
2. Name four Python data types.
3. What is the difference between a for loop and a while loop?
4. What is a function?
5. Write a Python program to print numbers from 1 to 5.

These topics provide a strong foundation for learning intermediate Python and AI development.